A Neuron Distribution in GPT2

The neuron distribution of GPT2-large is similar to Llama-7B, which is shown in Figure 9. Also, the curve of importance score in GPT2-large is similar to that in Llama-7B, illustrated in Figure 10.

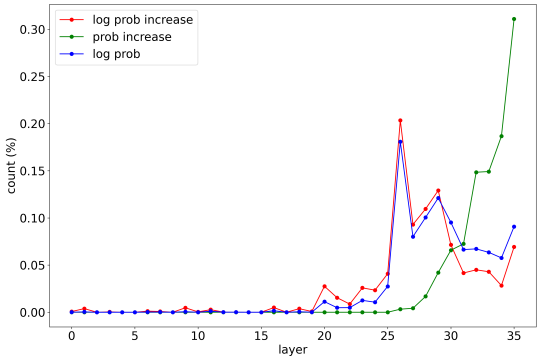

Figure 9: Neuron distribution on all layers in GPT2.

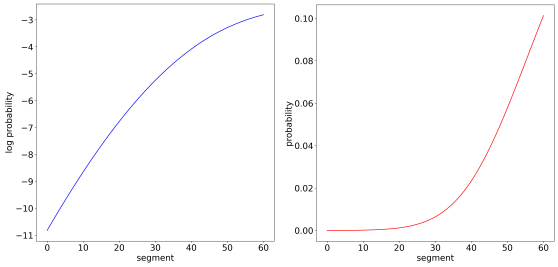

Figure 10: Curves of log probability increase (left) and probability increase (right) on GPT2.

B Head-Level Storage in GPT2/Llama

The top10 heads are shown in Table 10, where a_l^j is short for the jth head in lth attention layer. Knowledge with similar semantics is stored in the same heads (e.g. a_{30}^6 in GPT2 and a_{23}^{12} in Llama).

type	top10 heads
lang	$a_{30}^6, a_{26}^{17}, a_{26}^7, a_{32}^{11}, a_{19}^0, a_{31}^9, a_{25}^{13}, a_{22}^{17}, a_{28}^{13}, a_{29}^2$
capi	$a_{26}^7, a_{30}^6, a_{26}^{17}, a_{22}^{17}, a_{25}^{13}, a_{28}^{13}, a_{19}^0, a_{19}^{10}, a_{29}^2, a_{32}^{11}$
cnty	$a_{26}^7, a_{30}^6, a_{22}^{17}, a_{28}^{13}, a_{26}^{17}, a_{32}^{11}, a_{19}^0, a_{25}^{13}, a_{31}^9, a_{19}^{10}$
col	$a_{33}^5, a_{34}^1, a_{26}^7, a_{24}^{19}, a_{23}^{18}, a_{32}^{13}, a_{30}^1, a_{22}^8, a_{32}^{14}, a_{28}^2$
num	$a_{22}^{18}, a_{17}^3, a_{23}^8, a_{19}^2, a_{30}^3, a_{25}^{19}, a_{20}^3, a_{30}^0, a_{12}^2, a_{25}^3$
mon	$a_{27}^2, a_{26}^7, a_{25}^{11}, a_{19}^{10}, a_{30}^2, a_{28}^4, a_{23}^{18}, a_{17}^{17}, a_{33}^1, a_{17}^3$
lang	$a_{23}^{12}, a_{19}^{31}, a_{31}^{25}, a_{25}^{25}, a_{16}^5, a_{18}^1, a_{21}^9, a_{29}^{22}, a_{21}^{17}, a_{18}^{23}$
capi	$a_{23}^{12}, a_{29}^{22}, a_{25}^{25}, a_{31}^{25}, a_{19}^{31}, a_{18}^1, a_{16}^{15}, a_{16}^5, a_{21}^9, a_{18}^{23}$
cnty	$a_{23}^{12}, a_{19}^{31}, a_{25}^{25}, a_{21}^9, a_{31}^{25}, a_{16}^{15}, a_{18}^1, a_{16}^5, a_{29}^{22}, a_{28}^{19}$
col	$a_{29}^{22}, a_{28}^{19}, a_{20}^{27}, a_{16}^{15}, a_{17}^{27}, a_{28}^{21}, a_{25}^{14}, a_{18}^{28}, a_{24}^1, a_{14}^3$
num	$a_{28}^{19}, a_{26}^{24}, a_{23}^{10}, a_{30}^{13}, a_{21}^{29}, a_{13}^{24}, a_{18}^{24}, a_{29}^{22}, a_{17}^{23}, a_{19}^1$
mon	$a_{21}^{10}, a_{16}^0, a_{21}^{22}, a_{23}^{18}, a_{28}^{16}, a_{19}^{20}, a_{31}^6, a_{19}^1, a_{14}^3, a_{20}^{13}$

Table 10: Top10 important heads in GPT2 (first block) and Llama (second block).

The MRR decrease (%)/probability decrease (%) when intervening the top 1% heads for each knowledge is shown in Table 11. When intervening the top 1% heads for each knowledge, similar knowledge (language, capital and country) is affected a lot, while other knowledge (month, color, number) is not affected much.

	lang	capi	cnty	mon	col	num
lang	**44/51**	**33/52**	**38/56**	3/0	6/5	1/2
capi	**32/38**	**42/53**	**39/54**	15/11	18/12	0/1
cnty	**39/44**	**40/54**	**44/60**	3/0	10/5	2/3
mon	19/24	14/19	13/19	**55/63**	24/23	5/4
col	6/6	1/0	2/4	13/12	**49/59**	5/7
num	11/14	2/10	7/11	17/21	19/16	**33/34**
lang	**24/42**	**17/35**	**13/33**	0/0	6/15	1/1
capi	**38/58**	**28/50**	**22/53**	1/0	7/16	0/0
cnty	**42/61**	**31/54**	**28/58**	0/0	10/21	2/6
mon	7/14	4/11	7/13	**51/66**	8/13	3/8
col	3/12	6/16	6/14	13/25	**33/42**	1/10
num	0/0	1/4	1/2	2/9	3/13	**33/31**

Table 11: MRR decrease (%)/probability decrease (%) in GPT2 (first block) and Llama (second block) when intervening top 1% heads for different knowledge.

C Neuron-Level Storage in GPT2/Llama

The sum score of top neurons and all neurons in GPT2 are shown in Table 12. The sum importance score of top200 attention neurons and top100 FFN neurons are similar to those of all neurons.

	lang	capi	cnty	col	num	mon
all	7.3	6.7	6.6	3.6	2.3	4.3
positive	28.8	27.4	27.5	24.7	18.2	23.3
top100	4.0	3.7	3.5	2.7	1.6	2.5
top200	5.7	5.3	5.1	4.0	2.5	3.7
all	0.0	1.9	1.4	2.5	4.0	1.9
positive	70.1	69.4	69.1	72.0	62.3	66.6
top100	4.2	4.6	4.3	4.3	4.1	4.7
top200	6.0	6.3	5.9	6.1	5.7	6.4

Table 12: Imporatnce of top neurons in attention (first block) and FFN (second block) layers in GPT2.

The MRR decrease (%) / probability decrease (%) when intervening the top 200 attention neurons and top100 FFN neurons are shown in Table 13. The MRR score and probability score decreases around 91.1%/98.7% in GPT2, and 88.4%/97.1% in Llama. Therefore, our method can identify the important "value neurons" in both attention and FFN layers.

	lang	capi	cnty	col	num	mon
G	96/99	96/99	97/99	97/99	96/98	96/99
L	97/99	97/99	97/99	98/99	96/99	97/99

Table 13: MRR decrease (%) and probability decrease (%) when intervening the top200 attention neurons and top100 FFN neurons in GPT (G) and Llama (L).

D Important Query Layers for Attention Neurons in GPT2/Llama

We evaluate which layers have large inner product with top200 attention neurons, shown in Table 14. For every knowledge, the shallow and medium FFN layers play larger roles than attention layers.

	top10 query layers for attention neurons
lang	$f_0, f_1, a_0, f_2, f_{19}, f_{20}, f_3, f_{17}, f_{18}, f_{21}$
capi	$f_0, f_1, a_0, f_2, f_3, f_{20}, f_5, f_4, f_{19}, f_{17}$
cnty	$f_0, f_1, f_{19}, a_0, f_{18}, f_2, f_3, f_{21}, f_{20}, f_{17}$
col	$f_0, f_1, f_2, f_{23}, f_{20}, f_{21}, f_{22}, f_{24}, a_0, f_3$
num	$f_0, f_{18}, f_1, f_{19}, f_{22}, f_{16}, f_{21}, f_2, f_{12}, f_{20}$
mon	$f_0, f_1, f_{19}, f_2, f_9, f_{22}, f_{10}, f_{21}, a_0, f_{18}$
lang	$f_{20}, f_{19}, a_{16}, f_{16}, f_{15}, f_{18}, f_{14}, f_{21}, f_{12}, f_{21}$
capi	$f_{20}, f_{24}, f_{22}, f_{23}, f_{19}, a_{16}, a_{23}, f_{28}, f_{18}, f_{25}$
cnty	$f_{18}, f_{21}, f_{19}, a_{18}, f_{22}, a_{14}, a_{16}, f_{17}, a_{21}, a_{19}$
col	$f_{15}, f_{18}, f_{20}, f_{16}, f_{19}, f_{13}, f_{17}, f_{22}, f_{24}, f_{14}$
num	$f_{24}, f_{17}, f_{19}, f_{23}, f_{22}, f_{20}, f_{18}, f_2, f_{25}, f_{21}$
mon	$f_{14}, a_{16}, f_{19}, a_{19}, f_{18}, f_{20}, f_{13}, f_{17}, f_{15}, f_{22}$

Table 14: Top10 query layers for top200 attention neurons in GPT2 (first block) and Llama (second block).